

NAME

mmap2 – map files or devices into memory

LIBRARY

Standard C library (*libc*, *-lc*)

SYNOPSIS

```
#include <sys/mman.h> /* Definition of MAP_* and PROT_* constants */
#include <sys/syscall.h> /* Definition of SYS_* constants */
#include <unistd.h>

void *syscall(SYS_mmap2, unsigned long addr, unsigned long length,
              unsigned long prot, unsigned long flags,
              unsigned long fd, unsigned long pgof fset);
```

DESCRIPTION

This is probably not the system call that you are interested in; instead, see **mmap(2)**, which describes the glibc wrapper function that invokes this system call.

The **mmap2()** system call provides the same interface as **mmap(2)**, except that the final argument specifies the offset into the file in 4096-byte units (instead of bytes, as is done by **mmap(2)**). This enables applications that use a 32-bit *off_t* to map large files (up to 2⁴⁴ bytes).

RETURN VALUE

On success, **mmap2()** returns a pointer to the mapped area. On error, *-1* is returned and *errno* is set to indicate the error.

ERRORS**EFAULT**

Problem with getting the data from user space.

EINVAL

(Various platforms where the page size is not 4096 bytes.) *of fset * 4096* is not a multiple of the system page size.

mmap2() can also return any of the errors described in **mmap(2)**.

VERSIONS

On architectures where this system call is present, the glibc **mmap()** wrapper function invokes this system call rather than the **mmap(2)** system call.

This system call does not exist on x86-64.

On ia64, the unit for *offset* is actually the system page size, rather than 4096 bytes.

STANDARDS

Linux.

HISTORY

Linux 2.3.31.

SEE ALSO

getpagesize(2), **mmap(2)**, **mremap(2)**, **msync(2)**, **shm_open(3)**